

WEEK 7 PROJECT SUMMARY

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Track: Data Analytics

Project Phase: Week 7 (Testing & End-to-End Validation)

1. Testing Overview & Internal Validation

What Was Tested: The Week 6 Power BI "Advanced Insights" dashboard, specifically focusing on KPI accuracy, cross-filtering reliability, and the mathematical consistency of the logistical outcome tables.

Testing Results & Refinements: The dashboard perfectly reflected the 5,000 baseline appointment records without data loss. Slicer stress-testing (adjusting Gender and Age Group demographics) confirmed that visual components dynamically updated without breaking.

Key Finding Validated: The primary conclusion from Week 6—that clinic distance and waiting times do not drive no-shows—remained mathematically consistent across all tested demographic segments.

2. Cross-Track Collaboration & End-to-End Validation

Track Collaborated With: Data Science.

Collaborative Testing Activity: Evaluated the impact of the Data Analytics feature recommendations (booking_lead_days and reminder_channel) on the Data Science baseline classification model.

Outcome & Project Improvement: Testing revealed that the baseline model struggled with false negatives when relying solely on demographics. Retesting the model with the operational features generated by the Data Analytics track significantly improved the algorithm's predictive recall for no-shows.

3. Limitations, Challenges & Important Decisions

Decisions Made: Decided to finalize the dashboard without adding further complex visuals to maintain executive clarity and focus purely on booking lead times and reminder channels.

Remaining Limitations: The dataset cannot account for external human factors (e.g., sudden illness, traffic, or emergencies) that cause unpredictable no-shows, meaning the predictive model will always have a margin of error.

Major Challenges: Ensuring that the Data Science model correctly interpreted the operational weight of the "None" reminder category required careful alignment between the analytical findings and the machine learning pipeline.

4. Project Contribution & Week 8 Readiness

Overall Contribution: The Data Analytics track successfully redirected the HealthConnect project focus away from waiting-room logistics and toward actionable communication and advanced-booking interventions.

Week 8 Readiness: There are no unresolved dependencies or technical issues on the analytics side. The dashboard, verified findings, and documented cross-track validations are fully complete and ready for final project presentation and integration in Week 8.